

Theory of Automata (cs3005)

Sessional-I Exam

Date: November 1, 2025

Course Instructor(s)

Dr. Sobia Tariq, Dr. Tahir Ejaz, Dr. Aasim

Qureshi, Mr. Hamad ul Qudous, Ms. Saba Tariq,

Ms. Saba Sajjid, Ms. Rubab Anam

Total Time (Hrs.): 1

Total Marks: 35

Total Questions: 2

23L-0771

Roll No

BCS-5J

Section

M. Z

Student Signature

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Attempt all questions. Show all workings on the answer sheet and record the final answers in the designated spaces on the question paper. Marks will not be awarded for answers without workings or if the final answer is not written in the specified space on the question paper. Attach the answer booklet to this question paper.

CLO 3	CLO 3	
Question 1 (23)	Question 2 (12)	Total Marks (35)
23	12	35

CLO #: 3

Q1: Develop PDA for the following language.

[marks 5 + 18]

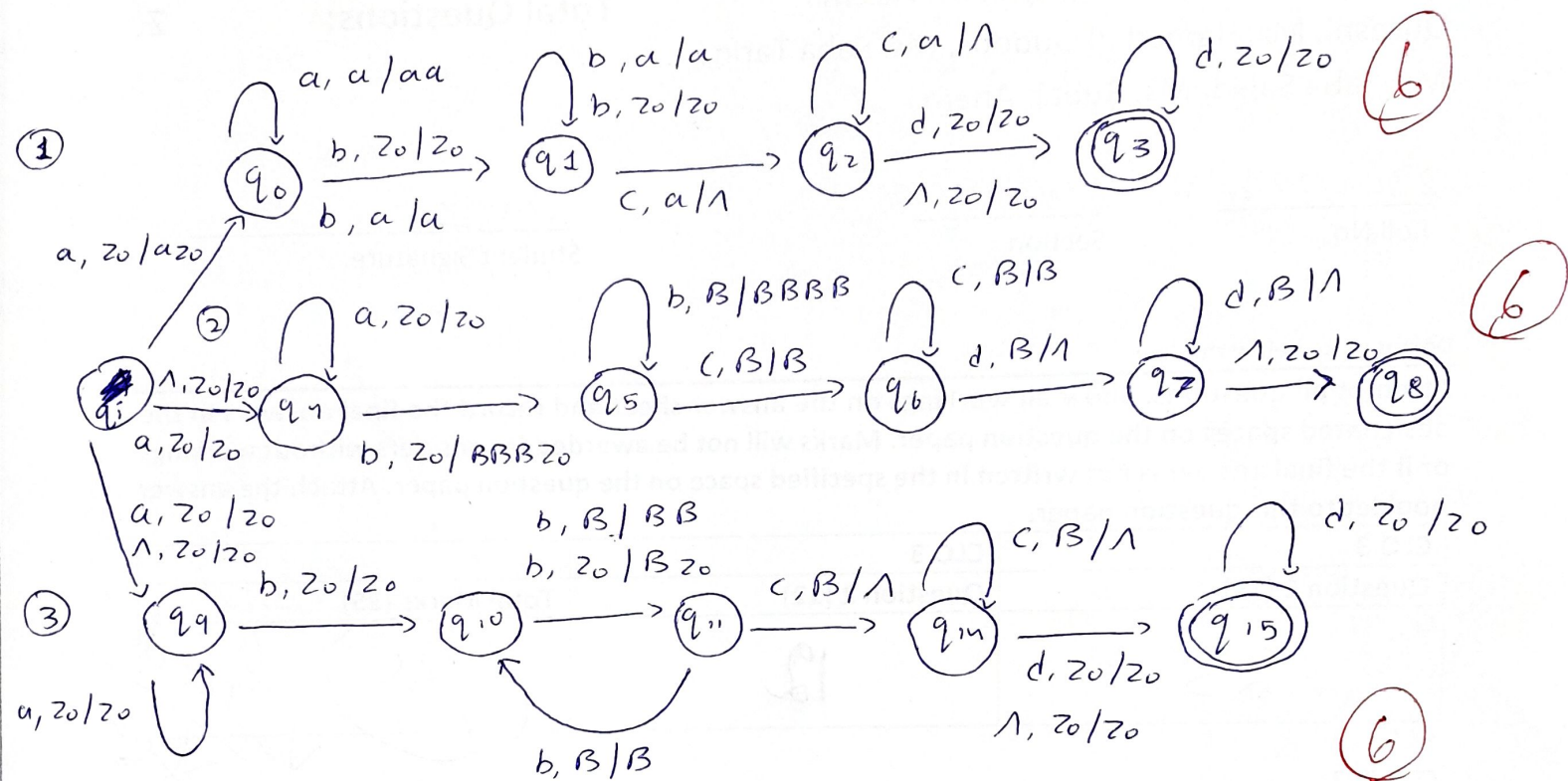
$$L = \{a^i b^j c^k d^l; i == k \text{ or } 3j == l \text{ or } j == 2k; i, j, k, l \geq 0\}$$

Enumerate at least 10 strings from the language in canonical order, ensuring that at least three strings are included for each case [$i=k$ or $3j=l$ or $j=2k$].

1. Λ (can be produced)	6. ($3j=l$) abddd
2. ($i=k$) ac	7. ($3j=l$) abccddd
3. ($i=k$) abc	8. ($j=2k$) bbcc
4. ($i=k$) abcc	9. abbbcc
5. ($3j=l$) bddd	10. abbbbcc

- ① handles $i == k$
- ② handles $3j == l$
- ③ handles $j == 2k$

I have made an NPDA to handle the multiple cursors.



rough work has been done on the answer sheet.

CLO #: 3

Q2: CYK on Robot Instructions

[10+2 marks]

 $\Sigma = \{\text{Left, Right, forward, turn, step, rotate}\}$

Consider each element in Σ as a single terminal. Left is 1st terminal, Right is 2nd terminal, and so on.
So there are a total of 6 terminals.

 $V = \{S, M, D, T, S_1, S_2\}$

A small robot understands this grammar in CNF form:

(start) $S \rightarrow MD \mid MT$
 $M \rightarrow \text{Left} \mid \text{Right}$
 $D \rightarrow S_1 D \mid \text{forward}$
 $T \rightarrow S_2 T \mid \text{turn}$
 $S_1 \rightarrow \text{step}$
 $S_2 \rightarrow \text{rotate}$

working has been done on the answer booklet

(Note: Grammar is already in CNF).

Apply the **CYK algorithm** to check if the instruction $x = \text{"Left step step step forward"}$ belongs to the Language generated by the above-mentioned grammar.

Note: "**Left step step step forward**" is a string with 5 terminals, as each element in Σ is a single terminal. Use only the required cells in the table given below (ignore the rest).

	5	4	3	2	1
1	S	ϕ	ϕ	ϕ	M
2	D	ϕ	ϕ	S_1	
3	D	ϕ	S_1		
4	D	S_1			
5	D				

the string is being processed as cell (1,5) contains an S which is the start symbol.

Accepted or not?

✓
Accept/Reject